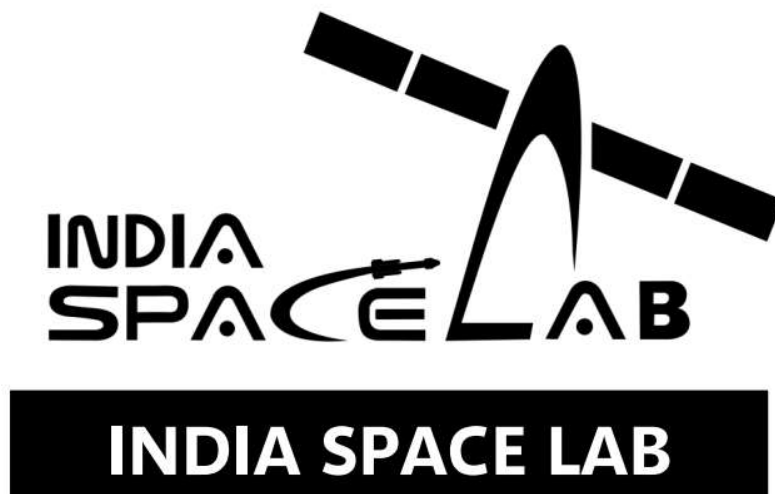


INTRODUCTION TO GUIDANCE AND CONTROL USING PYTHON SIMULATIONS



Project Report 01ST May to 15th June 2026

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INTRODUCTION

Guidance and Control systems are essential components of autonomous vehicles such as drones, rockets, spacecraft, and marine vehicles. These systems enable vehicles to maintain stability, follow predefined trajectories, and compensate for external disturbances.

This project was carried out using Python simulations executed in Google Colab. The simulations demonstrated the operation of PID controllers for drone altitude stabilization and feedback-based guidance algorithms for autonomous boat navigation.

OBJECTIVES

- Understand the fundamentals of guidance and control systems.
- Study PID controller tuning and performance.
- Observe the effects of environmental disturbances.
- Analyze trajectory tracking behaviour.
- Understand feedback correction mechanisms.
- Gain practical experience through Python-based simulations.

SOFTWARE AND TOOLS USED

- Google Colab
- Python Programming Language
- NumPy

- Matplotlib
- Interactive PID Controller Simulation
- Boat Guidance Simulation

PID CONTROLLER THEORY

PID stands for Proportional, Integral, and Derivative controller.

Proportional Control:

Responds directly to the current error.

Integral Control:

Eliminates accumulated error over time.

Derivative Control:

Predicts future error trends and improves stability.

A properly tuned PID controller provides:

- Fast response
- Low overshoot
- Reduced oscillations
- Improved disturbance rejection

DRONE ALTITUDE CONTROL SIMULATION

Aim:

To study drone altitude stabilization using a PID controller.

Procedure:

1. The simulation notebook was opened in Google Colab.
2. PID gains were adjusted using interactive sliders.
3. Wind disturbance was introduced after six seconds.
4. System response was observed and analyzed.

Parameters Used:

P Gain = 2.0

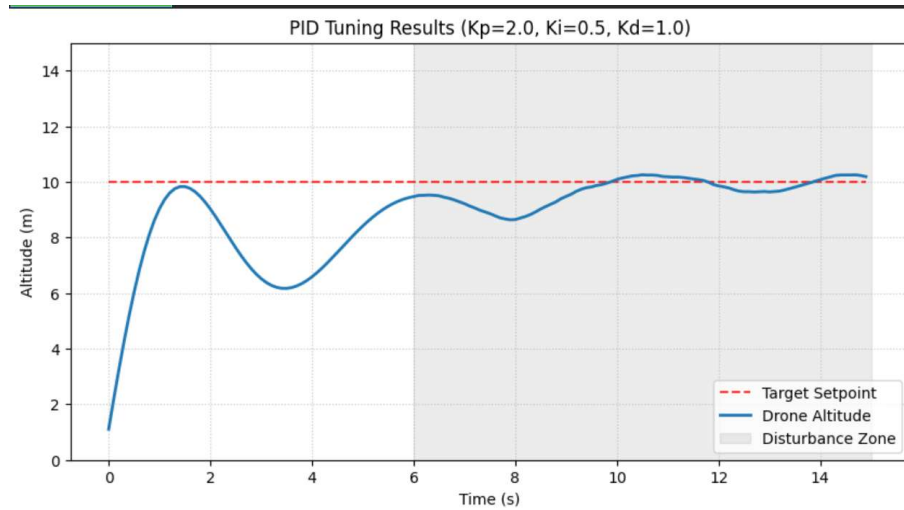
I Gain = 0.5

D Gain = 1.0

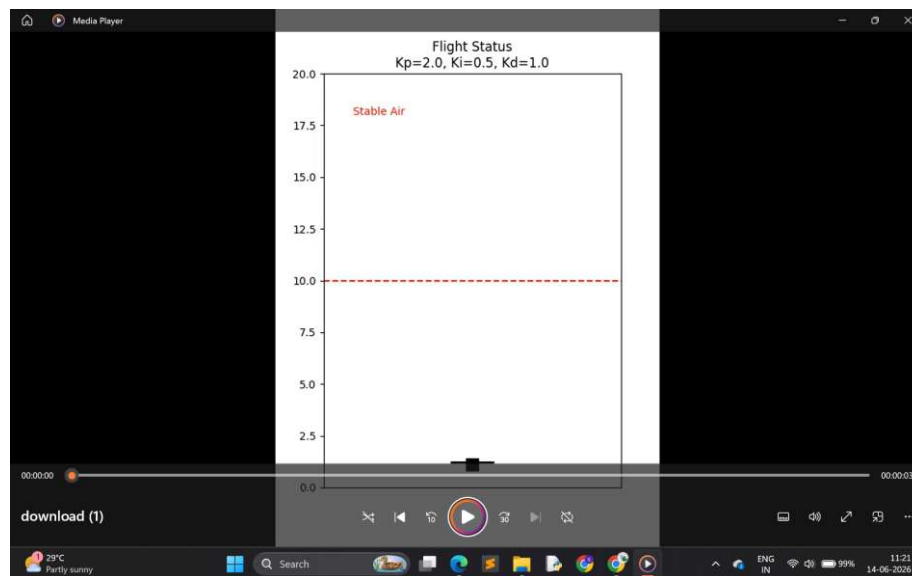
Wind Disturbance = 3

Observations:

- The drone successfully reached the target altitude.
- Small overshoot was observed initially.
- The controller stabilized the system rapidly.
- Wind disturbances caused temporary deviation.
- The drone recovered smoothly after disturbance.



Caption: Figure 1 – Drone Altitude Control Response Plot



Caption: Figure 2 – Drone Animation Output

BOAT GUIDANCE AND PATH TRACKING SIMULATION

Aim:

To study autonomous guidance and path-following behaviour.

Procedure:

1. The boat guidance simulation was executed.
2. Controller gains were adjusted.
3. Tracking performance was analyzed.
4. Simulations were performed under ideal conditions.

Parameters Used:

$K_P = 1.8$

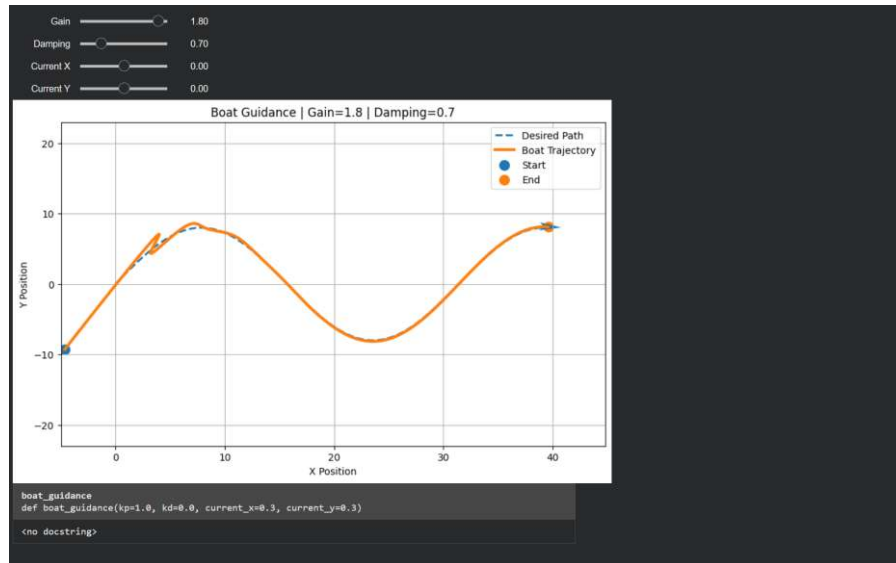
$K_D = 0.7$

Current $X = 0$

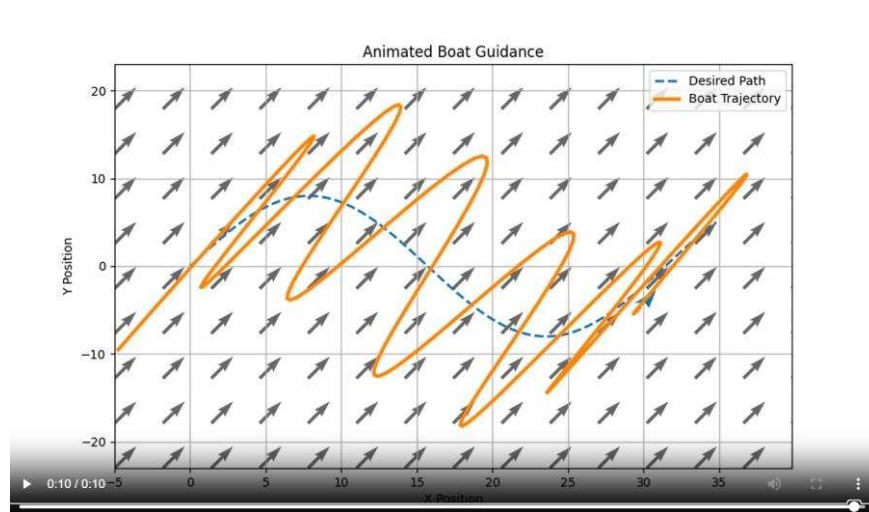
Current $Y = 0$

Observations:

- The boat followed the desired path accurately.
- Tracking error remained small.
- Motion was smooth and stable.
- Feedback correction improved navigation accuracy.



Caption: Figure 3 – Boat Path Tracking Graph



Caption: Figure 4 – Boat Guidance Animation

RESULTS AND OBSERVATIONS

Drone Simulation Results

- Stable altitude control achieved.
- Effective disturbance rejection observed.
- Minimal oscillation after tuning.
- Improved response characteristics.

Boat Simulation Results

- Accurate path tracking achieved.
- Stable navigation behaviour observed.
- Good tracking performance under ideal conditions.

Overall Results

The simulations successfully demonstrated the practical application of feedback control systems in autonomous vehicles.

DISCUSSION

The project demonstrated the importance of controller tuning in autonomous systems. Proper PID gain selection significantly improved stability and response quality.

The boat guidance simulation showed how feedback algorithms help vehicles remain on the desired trajectory despite environmental influences.

Both simulations highlighted real-world applications of guidance and control principles in aerospace and robotics engineering.

CONCLUSION

This project successfully demonstrated guidance and control concepts using Python simulations. PID control was applied to drone altitude stabilization, while feedback-based guidance algorithms were used for boat path tracking.

The simulations provided valuable insight into autonomous vehicle behaviour, controller tuning, and disturbance handling. These concepts form the foundation of modern drone and aerospace guidance systems.

REFERENCES

Python Documentation

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PID Control Theory

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